

Deadline: Feb 14

1. Let $gGDP_t$ denote the annual percentage change in gross domestic product and let int_t denote a short-term interest rate. Suppose that $gGDP_t$ is related to interest rate by

$$gGDP_t = \beta_0 + \beta_1 int_t + \beta_2 int_{t-1} + u_t,$$

where u_t is uncorrelated with int_t and int_{t-1} , and all other past values of interest rates.

Suppose that the central bank follows the monetary policy rule:

$$int_t = \gamma_0 + \gamma_1(1 - gGDP_{t-1}) + v_t,$$

where $\gamma_1 < 0$ (when last year's GDP growth is below 1 %, the central bank decreases interest rates to avoid a recession). If v_t is uncorrelated with all past values of int_t and u_t , prove that int_t must be correlated with u_{t-1} . In this case, which Gauss- Markov assumption is likely violated?

[Difficulty level: 3.0]

[Previous Final Exam Question]

2. 2. Okun's Law—see, for example, Mankiw (1994 Chapter 2) implies the following relationship between the annual percentage change in real GDP, $pcrgdp$, and the change in annual unemployment, $cunem$:

$$pcrgdp = 3 - 2 \times cunem.$$

If the unemployment rate is stable, real GDP grows at 3% annually. For each percentage point increase in the unemployment rate, real GDP grows by two percentage points less.

To see if the data on the U.S. economy support Okun's Law, we specify a model that allows deviations via an error term, $pcrgdp_t = \beta_0 + \beta_1 cunem_t + u_t$.

(a) Use Okun dataset to estimate the equation. Do you get exactly 3 for the intercept and -2 for the slope? Did you expect to?

(b) Based on the report in (a), find the t statistic for testing $H_0: \beta_1 = -2$. Do you reject H_0 against the two-sided alternative at any reasonable significance level?

- (c) Based on the report in (a), find the t statistic for testing $H_0: \beta_0 = 3$. Do you reject H_0 at the 5% significance level against the two-sided alternative at any reasonable significance level?
- (d) Find the F statistic and p-value for testing $H_0: \beta_0 = 3$ and $\beta_1 = -2$ against the alternative that H_0 is false. Does the test reject at the 10% level? Overall, would you say the data reject or tend to support Okun's law?
- (e) Are your conclusions for (a)-(d) reliable? Explain
- (f) In the regression $pcrgpd_t = \beta_0 + \beta_1 cunem_t + u_t$, please test the errors for $AR(1)$ serial correlation, without assuming $\{cunem_t: t = 1, 2, \dots\}$ is strictly exogenous. What do you conclude?
- (g) Regress squared residuals, $\hat{u}^2$, on $cunem_t$ (this is the Breush-Pagan test for heteroskedasticity in the simple regression case). What do you conclude?
- (h) Obtain the heteroskedasticity-robust standard error for the OLS estimate $\hat{\beta}_1$. Is it substantially different from the usual OLS standard error?

[Difficulty level: 3.0]

3. In an equation for quarterly data, suppose that $int_t = 1.6 + 0.48inf_t - 0.15inf_{t-1} + u_t$, where int is an interest rate and inf is the inflation rate. What is the impact propensity? What is the cumulative effect?

[Difficulty level: 1.5]